A-A-53271
30 November 1987

COMMERCIAL ITEM DESCRIPTION

ARTICULATOR, DENTAL (Semi-Adjustable, Arcon Style) and PLATE, MOUNTING, DENTAL ARTICULATOR

The General Services Administration has authorized the use of this Commercial Item Description.

This commercial item description covers a Hanau type, semi-adjustable articulator and a pair of mounting plates (Replacement).

Class 1. Articulator, Complete

Class 2. Plate, Mounting (Replacement)

Salient characteristics:

Class 1 anatomical articulator shall be a Hanau type, Arcon style, complete dental articulator and shall be suitable for use in the establishment of correct relationships and occlusion during the fabrication of dental prosthesis, including edentulous cases. Class 2, replacement dental mounting plate, shall consist of a pair of dental articulator mounting plates as replacement plates suitable for use with Class 1 articulators.

The articulator shall be designed to permit precise:

(1) Establishment and locking in of centric relation.

(2) Adjustment of condylar guide elements for the reproduction of protrusive and lateral mandibular excursions.

(3) Adjustment of incisal guide elements for guidance of the incisal pin.

The articulator shall conform to DPSC Drawing number 24557. All mating parts shall fit properly. All hinges shall provide free operation and movement of the hinged parts. All locking screws and thumbnut shall lock the parts securely in the required position.

AMSC N/A FSC 6520

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Shall be capable of accepting a face bow (NSN 6520-00-935-5826) transfer of the maxillary arch and shall provide an orbitale indicator for relating the patient's infra-orbitale notch as a third anatomical mounting reference.

Shall have condylar guidances which shall be the control center of the articulator. The guidances shall adjustably assimilate the multiple functions of the glenoid fossa from centric to protrusive, or from centric to a right or left lateral relation. The outside diameter of the condylar guidances shall accurately rotate inside the horizontal bearing surface of the condylar post.

An enclosed blind end track shall be provided in the diametrical edge of the condylar guidance to captively retain the slidable and rotatable spherical condylar element. A blind stop at the blind end of the enclosed track shall limit the anterior movement of the condylar element.

When the condylar element rests against this stop, it shall be at the centric position. A threaded stud shall be fixed to the outer diameter of the condylar guidance and shall bear in a keyway of the condylar post. The condylar guidance shall rotate freely without lateral side shift in the condylar post. An indicating line shall appear at the outer edge of the condylar guidance, opposite to the enclosed track. When this line is aligned with the "0" in the condylar post, the enclosed track for the condylar element shall be at a zero degree horizontal position. The enclosed condylar track inclination shall be adjustable from zero to 60 degrees on the plus side and from zero to 20 degrees on the minus side. Each calibrated line on the condylar post shall equal 5 angular degrees of inclination. The adjustment of this inclination shall be maintained by engaging a thumbnut.

The condylar posts shall rotate on vertical axis. A threaded axial rod shall enter into the hole in the upper member and shall form a fixed intercondylar width of 11 centimeters with the other condylar post. A spring shall be placed over the axial rod and shall be enclosed by a thumbnut to assure a constant vertical tension on the condylar post during adjustment. An indicating line shall appear on the upper member and shall abut the calibrations on the flange of the right and left pads affixed to the condylar post. When this line is aligned with the "0" on the flange of the condylar post, the enclosed track for the condylar element shall be parallel with the sagittal plane.

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The calibrations on the flange of the pads shall read from zero to 30 degrees, each line equaling 5 angular degrees of lateral inclination. The adjustment of this lateral inclination shall be retained by tightening the thumbnut.

The solid condylar shafts shall adjustably slide in the right and left wings of the lower member and shall act as axles for the condylar elements. The shafts shall be axially adjusted to contact their shouldered ends with the flats on the sides of the condylar elements. A side shift of less than .003 inch of the upper member shall exist so that the articulator will not bind at a centric position. The adjustment of the condylar shafts shall be by means of setscrews. The outboard ends of the shafts shall be formed to accept the conventional facia type face bow.

A rollpin shall be extended from the right and left wings of the lower member. The rollpin shall be 12 millimeters behind and on the same horizontal plane as the condylar shaft. The pins shall suspend the earpieces of an ear bow and shall simulate the external auditory meatus.

An incisal pin shall serve as the forward control of the articulator. The pin shall maintain a vertical stop and provide a stylus contact for the excursive movements of the articulator against the various inclined guiding surfaces of the incisal guide. The flat side of the incisal pin shall have a line in the middle, and five additonal lines which shall be calibrated in millimeters and shall extend on either side of the middle line. The incisal pin shall be inserted into the upper member and locked in position by means of a thumbscrew. The spade-shaped end of the pin shall be 90 degress to, and in contact with, the center of the central table on the incisal guide, and shall provide a parallelism of the upper to the lower member. An incisal edge reference notched groove shall be present on the incisal pin, 47 millimeters below the condylar plane. The groove shall form an average vertical landmark for alignment of the incisal edge of the maxillary centrals, when making a face bow transfer without an orbitale pointer reference of the patient's infra-orbitale notch. An additional reference groove, 30 millimeters below the condylar plane, shall represent the incisal edge at the flat Bonwil plane. The incisal pin shall extend above the upper member to provide a third point of stability when inverting the articulator. The lower member surface finish shall include an anterior slot at its front to accept a sliding platform. This platform shall allow the adjustable incisal guide to slide forward or rearward to align its center axis with the spade shaped end of the incisal pin to prevent the opening or closing of the vertical dimension when adjusting the adjustable incisal guide.

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The square end of the incisal pin shall be spherically formed to provide a secondary function. The incisal pin shall allow the spherical end to be used in a custom fabrication of an acrylic incisal guidance when reversing it in the upper member. A plastic bumper shall be attached to the side of the upper member with a substantial steel screw. The bumper shall act as a shock absorber stop against the lower member wing when swinging the upper member open.

A centric lock shall be provided at the side of the condylar guidance for securing the condylar element in a centric position. When engaging the centric lock, an inclined plane shall depress a small centric pin in the condylar guidance track to arrest the condylar element. In this locked position, the upper member shall be restricted to opening and closing movements only. The releasing of the centric lock two full turns shall allow the centric pin to release and return the freedom of movement to the condylar element.

An incisal guide shall provide for an independent adjustment of a guiding incisal surface. The guide shall operate with the incisal pin and condylar guidances to result in a stable, three-dimensional, programmed, guide pattern for the mounted casts. The rocker of the guide shall rest in a rocker base fixed to the lower member. The rocker shall be retained in the base by a "T" shaped bolt which is adjustably fixed by a locknut. The rocker shall rotate antero-posteriorly in the rocker base from a horizontal zero degree to a 60 degree positive inclination of protrusion and to a negative of 20 degrees.

The central table of the rocker shall be 7/32 inch wide and shall form the inclined surface for the protrusion of the incisal pin. Lateral wings shall be separately adjustable by means of elevating screws from a zero horizontal to a 45 degree angulation reading as shown on the calibrations present on the rocker.

An orbitale indicator shall be present at 7 millimeters above the articulator's condylar plane. The indicator shall consist of a crescent shaped stamping and spacer attached to the upper member by means of a screw. The indicator shall rotate about a vertical axis and fixed in place by means of a thumbnut.

Mounting plates provided shall be suitable for luting the maxillary and mandibular casts to the upper and lower members of the articulator with gypsum material. The plates shall contain two, elongated and tapered luting slots for secure adherence of the gypsum mounting. The mounting plates shall be fabricated from zinc alloy.

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The plates shall have a brass domed insert at the center which shall be internally threaded (5/16 - 24 threads per inch) for attachment to the upper or lower member of the articulator. The plates shall have a keyway slot and a dowel hole which shall register over two dowels in the upper and lower member. Each mounting plate shall be round. The dowels shall insure the accurate attachment of the plates by means of thumbscrews. The thumbscrews shall be removable from the upper and lower member by simultaneously pulling and unthreading the thumbscrews. In addition to the two pairs of mounting plates furnished with each articulator, two (2) extra pairs of mounting plates shall be supplied.

The aluminum upper and lower members shall be protected with a corrosion-inhibiting, baked, black teflon. The aluminum condylar posts, lateral wings, and orbitale indicator shall have a smooth, satin finish. The aluminum screws and nuts, namely DPSC Drawing 24557 Item Numbers 12, 13, 24, 27, 31, 36 and 37 shall have a smooth, black anodized finish. The aluminum incisal pin and all aluminum screws and nuts shall have a smooth, black anodized finish. The zinc alloy mounting plates shall have a tumbled finish. The remaining aluminum screws or nuts shall have a smooth, satin aluminum finish. The aluminum incisal pin 38 shall have black anodized ends and lines, contrasting against its smooth, satin finished diameter. Thumbscrews, thumbnuts, ends of the incisal pin, and lines of calibration shall be black anodized.

The articulator shall be furnished with a booklet containing adequate instructions for operational use and maintenance.

Workmanship. The articulator shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One articulator, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one articulator. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, AQL 2.5, and sampling for tests (when applicable) shall be conducted at inspection level S-1, AQL 1.0.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

National Stock Number	ITEM IDENTIFICATION	DESIGNATION
6520-01-108-4850	ARTICULATOR, DENTAL (Semi-Adjustable, Arcon Style).	CLASS 1
6520-01-113-4746	PLATE, MOUNTING, DENTAL, ARTICULATOR	CLASS 2

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
PHS
FDA-MPQAS

49108

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

Project No. 6520-2014

Location: NCR CID DISK 3

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53271	ARTICULATOR, DENTAL (Semi-Adjustable, Arcon Style) and PLATE, MOUNTING, DENTAL ARTICULATOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* - Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* - Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* - Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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